

WIA1006/WID3006

MACHINE LEARNING

Semester 2, Session 2025/2026 | Universiti Malaya

Home of the Bright, Land of the Brave
Di Sini Bermulanya Pintar, Tanah Tumpahnya Berani

www.um.edu.my



UNIVERSITI
MALAYA

SUPPORT VECTOR MACHINE (SVM)

Home of the Bright, Land of the Brave
Di Sini Bermulanya Pintar, Tanah Tumpahnya Berani



www.um.edu.my



[universityofmalaya](https://www.facebook.com/universityofmalaya)



[unimalaya](https://www.instagram.com/unimalaya)



[uniofmalaya](https://www.youtube.com/uniofmalaya)



UNIVERSITI
MALAYA

Today: Support Vector Machine (SVM)



A classifier derived from statistical learning theory by Vapnik, et al. in 1992



SVM became famous when, using images as input, it gave accuracy comparable to neural-network with hand-designed features in a handwriting recognition task



Currently, SVM is widely used in object detection & recognition, content-based image retrieval, text recognition, biometrics, speech recognition, etc.



Also used for regression (will not cover today)

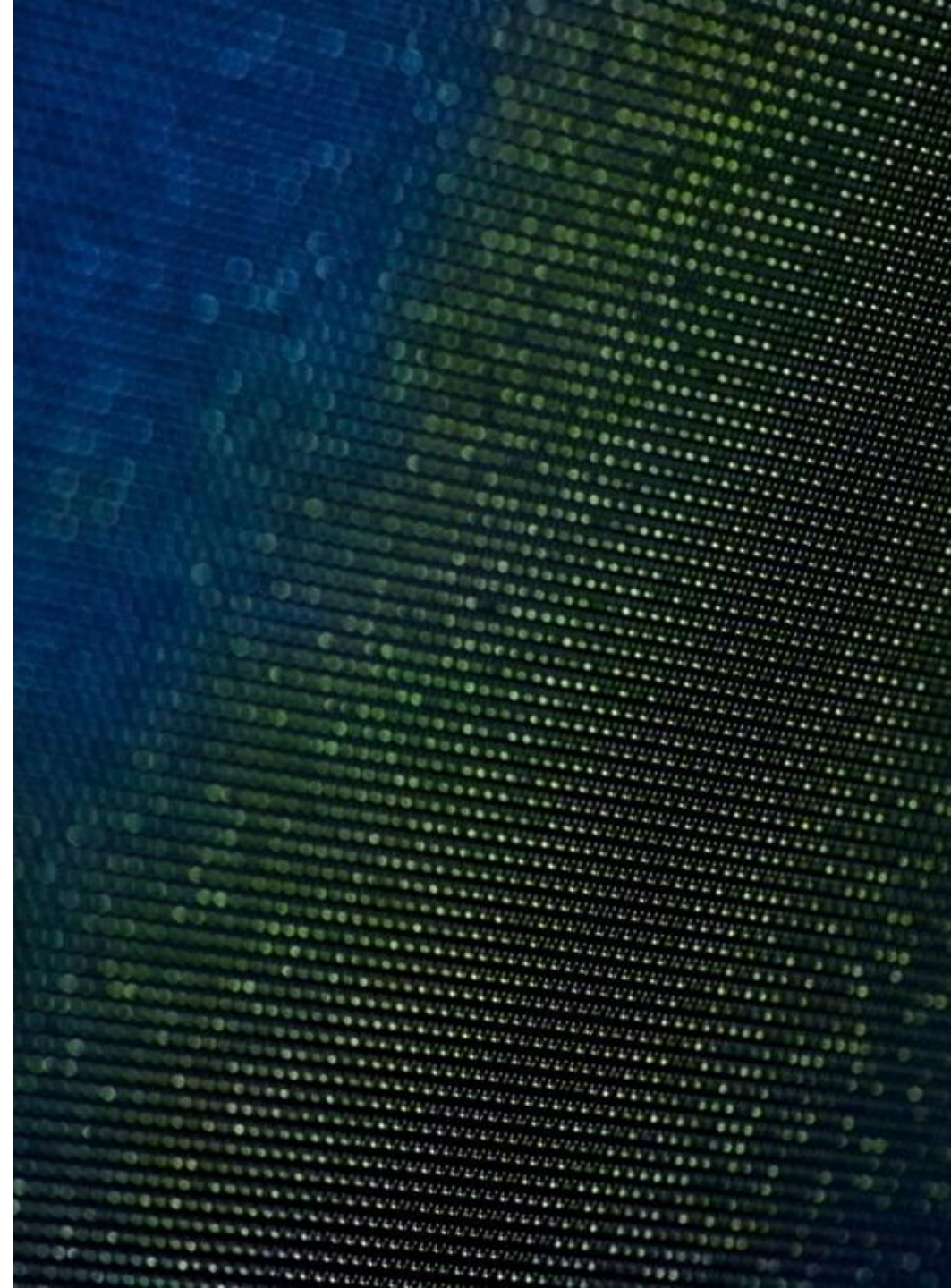
Outline

Linear Discriminant Function

Large Margin Linear Classifier

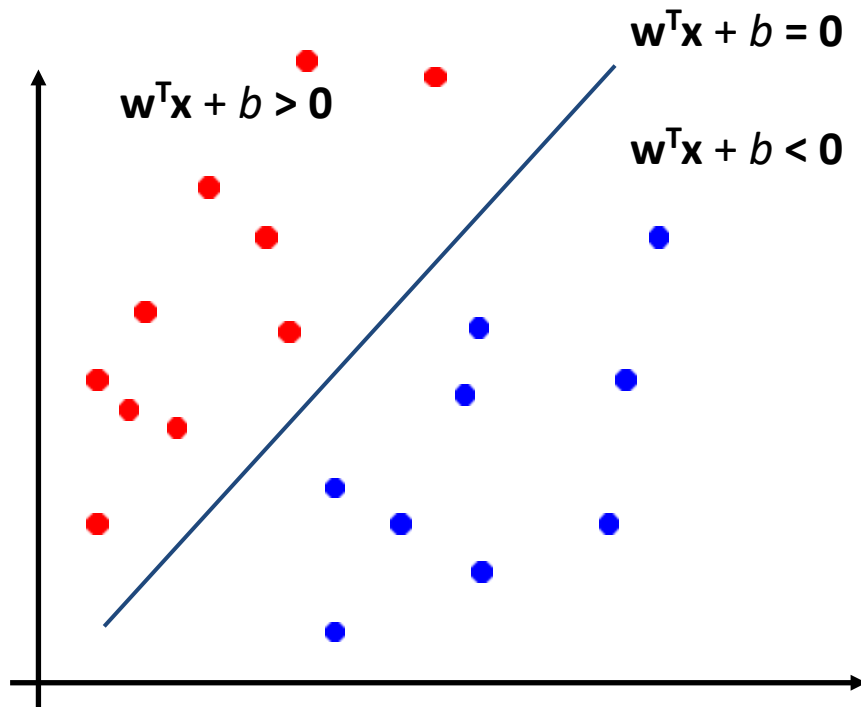
Nonlinear SVM: The Kernel Trick

Demo of SVM



Perceptron Revisited: Linear Separators

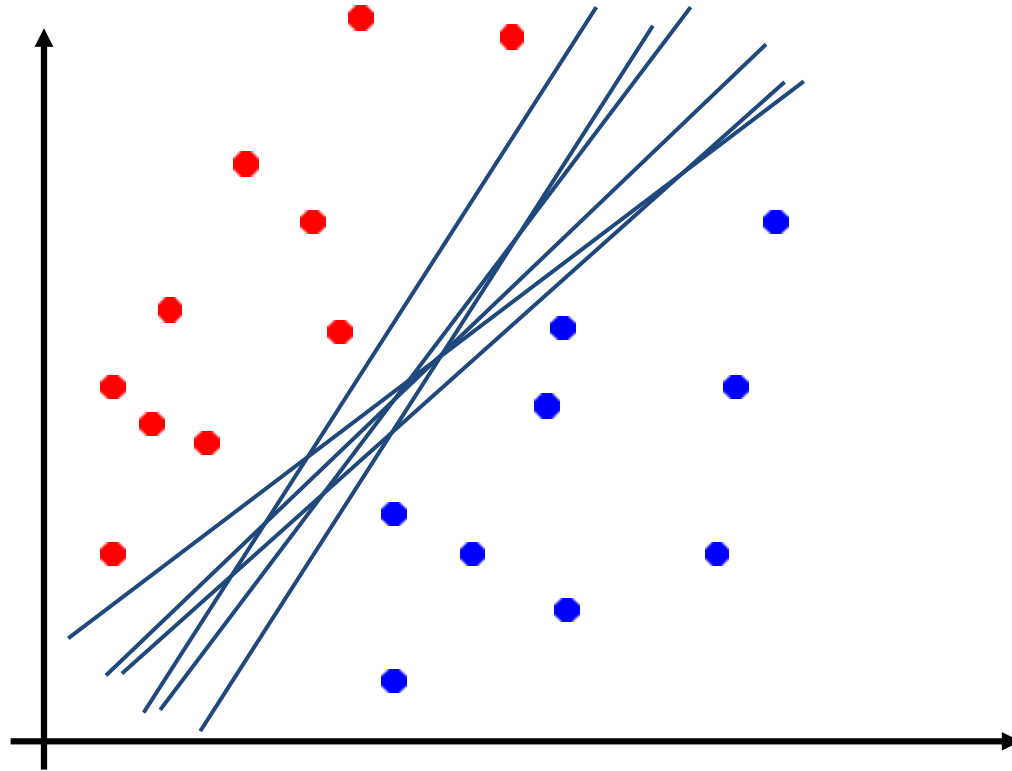
- Binary classification can be viewed as the task of separating classes in feature space:



$$f(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x} + b)$$

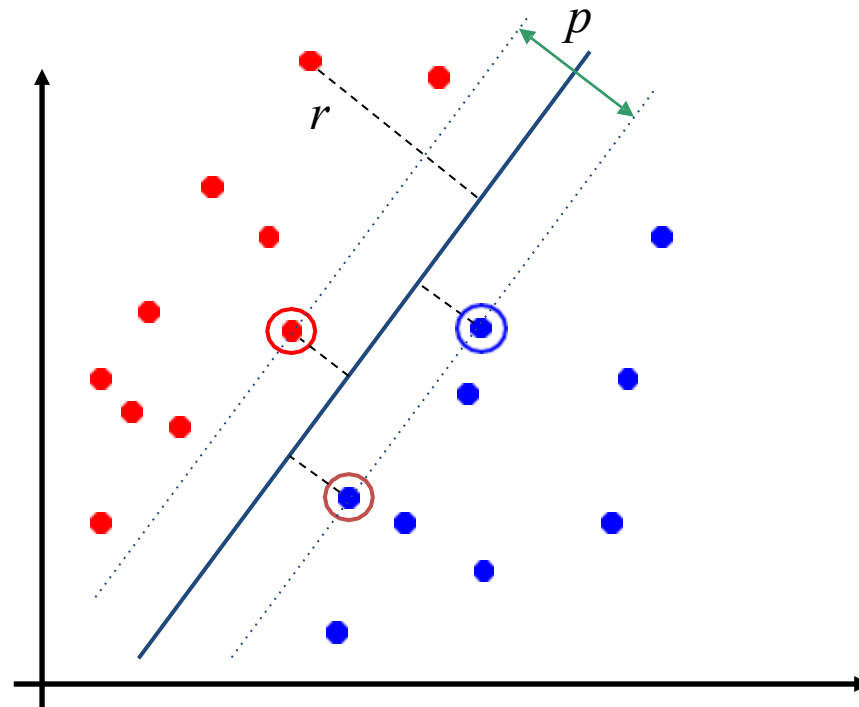
Linear Separators

- Which of the linear separators is optimal?



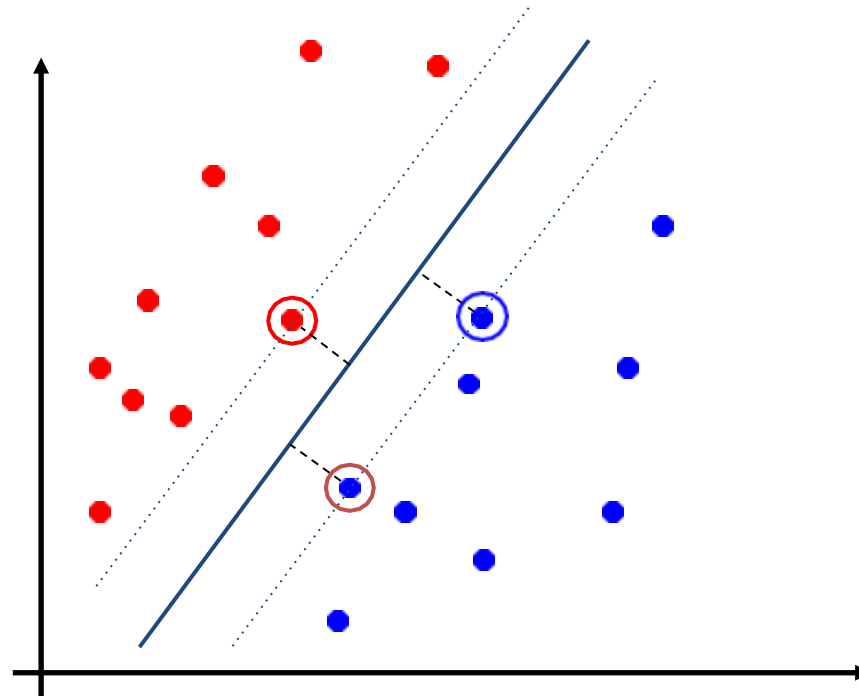
Classification Margin

- Distance from example $\mathbf{x}$ to the separator is $r = \frac{\mathbf{w}^T \mathbf{x} + b}{\|\mathbf{w}\|}$
- Examples closest to the hyperplane are **support vectors**.
- **Margin** ρ of the separator is the distance between support vectors.



Maximum Margin Classification

- Maximizing the margin is good according to intuition and PAC theory.
- Implies that only support vectors matter; other training examples are ignorable.



Linear SVM Mathematically

- Let training set $\{(\mathbf{x}_i, y_i)\}_{i=1..n}$, $\mathbf{x}_i \in \mathbf{R}^d$, $y_i \in \{-1, 1\}$ be separated by a hyperplane with margin ρ . Then for each training example $(\mathbf{x}_i, y_i)$:

$$\begin{aligned} \mathbf{w}^T \mathbf{x}_i + b &\leq -\rho/2 & \text{if } y_i = -1 \\ \mathbf{w}^T \mathbf{x}_i + b &\geq \rho/2 & \text{if } y_i = 1 \end{aligned} \quad \Leftrightarrow \quad y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq \rho/2$$

- For every support vector $\mathbf{x}_s$ the above inequality is an equality. After rescaling $\mathbf{w}$ and b by $\rho/2$ in the equality, we obtain that distance between each $\mathbf{x}_s$ and the hyperplane is $r = \frac{y_s(\mathbf{w}^T \mathbf{x}_s + b)}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}$
- Then the margin can be expressed through (rescaled) $\mathbf{w}$ and b as:

$$p = 2r = \frac{2}{\|\mathbf{w}\|}$$

Linear SVM Mathematically

- Then we can formulate the *quadratic optimization problem*:

Find $\mathbf{w}$ and b such that
 $\rho = \frac{2}{\|\mathbf{w}\|}$ is maximized
and for all $(\mathbf{x}_i, y_i), i=1..n : y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1$

- Which can be reformulated as:

Find $\mathbf{w}$ and b such that
 $\Phi(\mathbf{w}) = \|\mathbf{w}\|^2 = \mathbf{w}^T \mathbf{w}$ is minimized
and for all $(\mathbf{x}_i, y_i), i=1..n : y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1$

Solving the Optimization Problem

Find $\mathbf{w}$ and b such that
 $\Phi(\mathbf{w}) = \mathbf{w}^T \mathbf{w}$ is minimized
and for all $(\mathbf{x}_i, y_i), i=1..n$: $y_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$

- Need to optimize a *quadratic* function subject to *linear* constraints.
- Quadratic optimization problems are a well-known class of mathematical programming problems for which several (non-trivial) algorithms exist.
- The solution involves constructing a *dual problem* where a *Lagrange multiplier* α_i is associated with every inequality constraint in the primal (original) problem:

Find $\alpha_1 \dots \alpha_n$ such that
 $\mathbf{Q}(\alpha) = \sum \alpha_i - \frac{1}{2} \sum \sum \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$ is maximized and
(1) $\sum \alpha_i y_i = 0$
(2) $\alpha_i \geq 0$ for all α_i

The Optimization Problem Solution

- Given a solution $\alpha_1 \dots \alpha_n$ to the dual problem, solution to the primal is:

$$\mathbf{w} = \sum \alpha_i y_i \mathbf{x}_i \quad b = y_k - \sum \alpha_i y_i \mathbf{x}_i^T \mathbf{x}_k \quad \text{for any } \alpha_k > 0$$

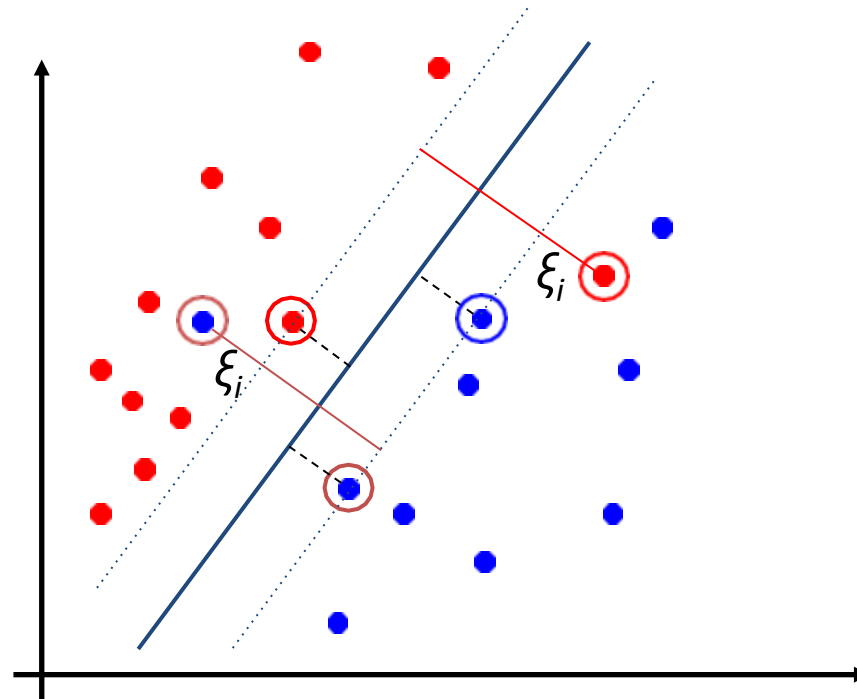
- Each non-zero α_i indicates that corresponding $\mathbf{x}_i$ is a support vector.
- Then the classifying function is (note that we don't need $\mathbf{w}$ explicitly):

$$f(\mathbf{x}) = \sum \alpha_i y_i \mathbf{x}_i^T \mathbf{x} + b$$

- Notice that it relies on an *inner product* between the test point $\mathbf{x}$ and the support vectors $\mathbf{x}_i$ — we will return to this later.
- Also keep in mind that solving the optimization problem involved computing the inner products $\mathbf{x}^T \mathbf{x}$ between all training points.

Soft Margin Classification

- What if the training set is not linearly separable?
- *Slack variables* ξ_i can be added to allow misclassification of difficult or noisy examples, resulting margin called *soft*.



Soft Margin Classification Mathematically

The old formulation:

Find $\mathbf{w}$ and b such that
 $\Phi(\mathbf{w}) = \mathbf{w}^T \mathbf{w}$ is minimized
and for all $(\mathbf{x}_i, y_i), i=1..n$: $y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1$

Modified formulation
incorporates slack
variables:

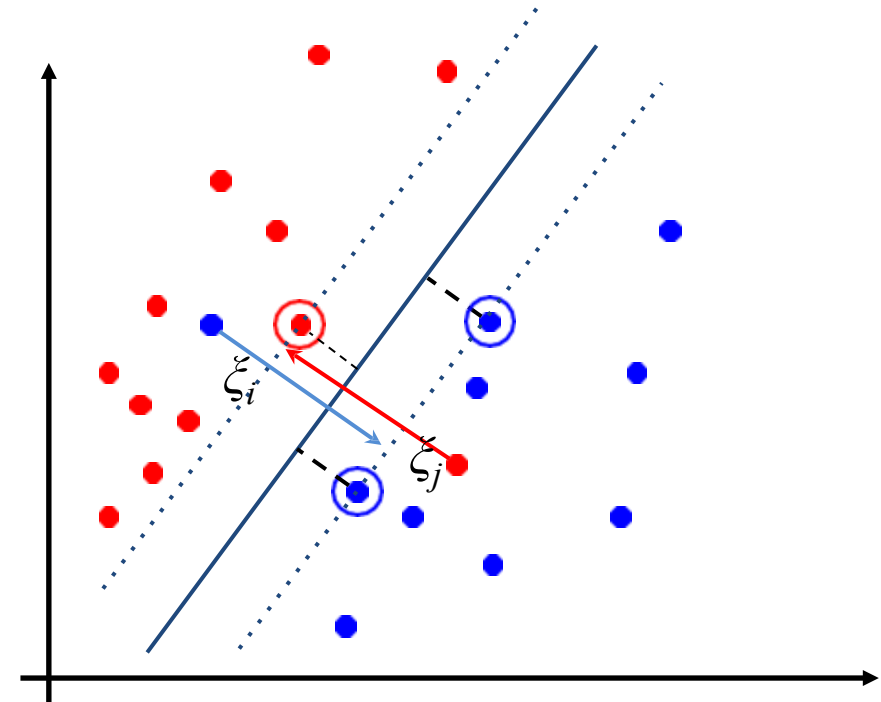
Find $\mathbf{w}$ and b such that
 $\Phi(\mathbf{w}) = \mathbf{w}^T \mathbf{w} + C \sum \xi_i$ is minimized
and for all $(\mathbf{x}_i, y_i), i=1..n$: $y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$

Parameter C can be
viewed as a way to control
overfitting:

- it “trades off” the relative importance of maximizing the margin and fitting the training data.

Soft Margin Classification

- If the training data is not linearly separable, *slack variables* ξ_i can be added to allow misclassification of difficult or noisy examples.
- **Allow some errors**
 - Let some points be moved to where they belong, at a cost
- Still, try to minimize training set errors, and to place hyperplane “far” from each class (large margin)



Soft Margin Classification – Solution

- Dual problem is identical to separable case (would *not* be identical if the 2- norm penalty for slack variables $C \sum \xi_i^2$ was used in primal objective, we would need additional Lagrange multipliers for slack variables):

Find $\alpha_1 \dots \alpha_N$ such that

$Q(\alpha) = \sum \alpha_i - \frac{1}{2} \sum \sum \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$ is maximized and

(1) $\sum \alpha_i y_i = 0$

(2) $0 \leq \alpha_i \leq C$ for all α_i

- Again, $\mathbf{x}_i$ with non-zero α_i will be support vectors.
- Solution to the dual problem is:

$$\mathbf{w} = \sum \alpha_i y_i \mathbf{x}_i$$

$$b = y_k (1 - \xi_k) - \sum \alpha_i y_i \mathbf{x}_i^T \mathbf{x}_k \quad \text{for any } k \text{ s.t.}$$

$$\alpha_k > 0$$

Soft Margin Classification – Solution (cont.)

- Again, we don't need to compute $\mathbf{w}$ explicitly for classification:

$$f(\mathbf{x}) = \sum \alpha_i y_i \mathbf{x}_i \cdot \mathbf{x} + b$$

Theoretical Justification for Maximum Margins

- Vapnik has proved the following:
 - *The class of optimal linear separators has VC dimension h bounded from above as*

$$h \leq \min \left\{ \left\lceil \frac{D^2}{\rho^2} \right\rceil, m_0 \right\} + 1$$

where ρ is the margin, D is the diameter of the smallest sphere that can enclose all of the training examples, and m_0 is the dimensionality.

- Intuitively, this implies that regardless of dimensionality m_0 we can minimize the VC dimension by maximizing the margin ρ .
- Thus, complexity of the classifier is kept small regardless of dimensionality.

Linear SVMs: Overview

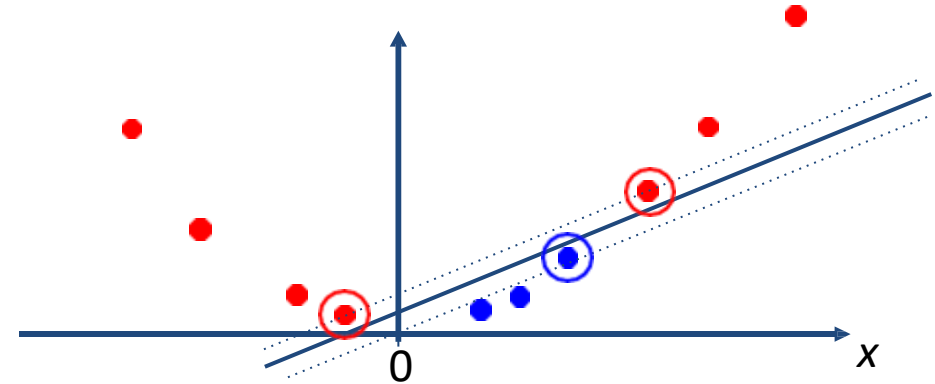
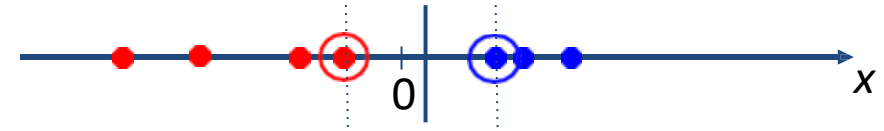
- The classifier is a *separating hyperplane*.
- Most “important” training points are support vectors; they define the hyperplane.
- Quadratic optimization algorithms can identify which training points $\mathbf{x}_i$ are support vectors with non-zero Lagrangian multipliers α_i .
- Both in the dual formulation of the problem and in the solution training points appear only inside inner products:

Find $\alpha_1 \dots \alpha_N$ such that
 $Q(\boldsymbol{\alpha}) = \sum \alpha_i - \frac{1}{2} \sum \sum \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$ is maximized and
(1) $\sum \alpha_i y_i = 0$
(2) $0 \leq \alpha_i \leq C$ for all α_i

$$f(\mathbf{x}) = \sum \alpha_i y_i \mathbf{x}^T \mathbf{x}_i + b$$

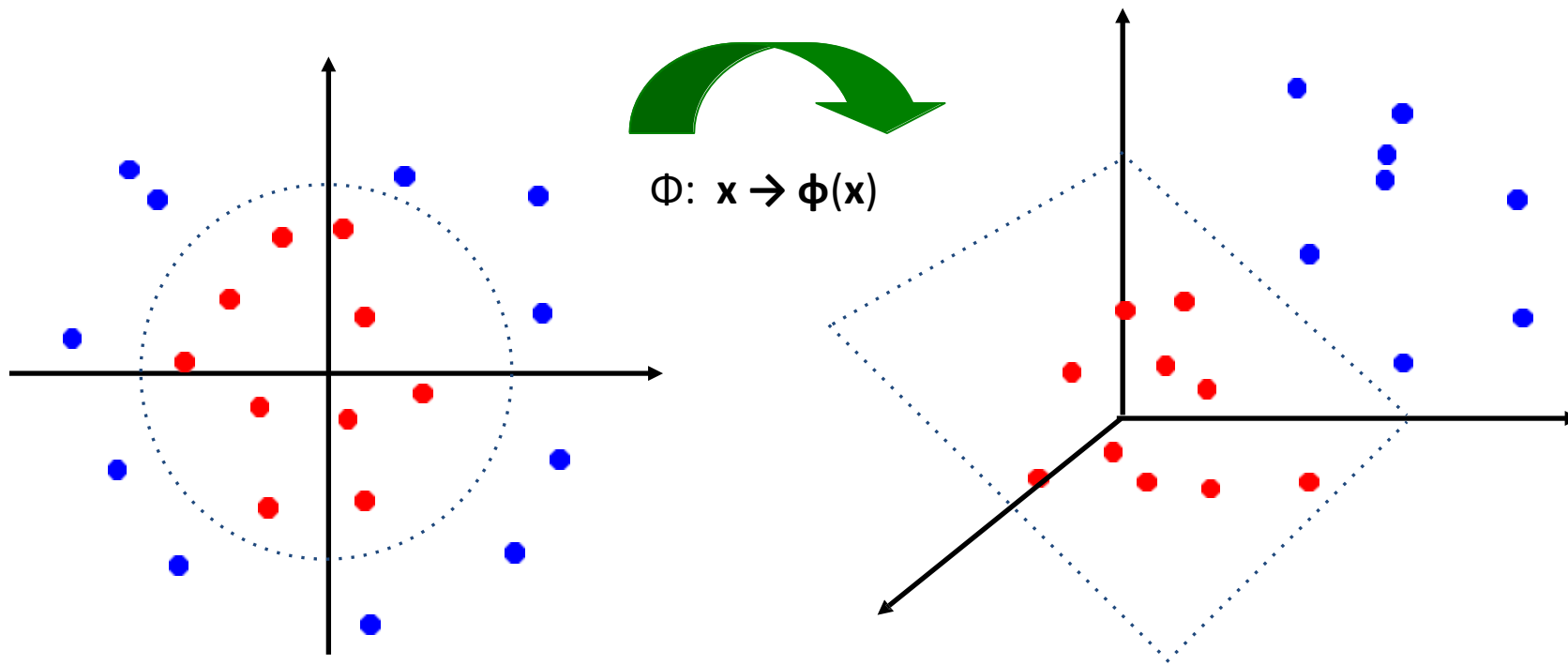
Non-linear SVMs

- Datasets that are linearly separable with some noise work out great:
- But what are we going to do if the dataset is just too hard?
- How about... mapping data to a higher-dimensional space:



Non-linear SVMs: Feature Spaces

- General idea: the original feature space can always be mapped to some higher-dimensional feature space where the training set is separable:



The “Kernel Trick”

- The linear classifier relies on inner product between vectors

$$K(x_i, x_j) = x_i^T x_j$$

- If every datapoint is mapped into high-dimensional space via some transformation $\phi: x \rightarrow \phi(x)$, the inner product becomes:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

- A *kernel function* is a function that is equivalent to an inner product in some feature space. Example:

2-dimensional vectors $x = [x_1 x_2]$; let $K(x_i, x_j) = (1 + x_i^T x_j)^2$

Need to show that $K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$

The “Kernel Trick” (cont.)

- $$\begin{aligned} K(x_i, x_j) &= (1 + x_i^T x_j)^2 = 1 + x_{i1}^2 x_{j1}^2 + 2x_{i1}x_{j1}x_{i2}x_{j2} + x_{i2}^2 x_{j2}^2 + 2x_{i1}x_{j1} + 2x_{i2}x_{j2} \\ &= [1 \ x_{i1}^2 \ \sqrt{2}x_{i1}x_{i2} \ x_{i2}^2 \ \sqrt{2}x_{i1} \ \sqrt{2}x_{i2}]^T [1 \ x_{j1}^2 \ \sqrt{2}x_{j1}x_{j2} \ x_{j2}^2 \ \sqrt{2}x_{j1} \ \sqrt{2}x_{j2}] \\ &= \phi(x_i)^T \phi(x_j) , \text{ where } \phi(x) = [1 \ x_1^2 \ \sqrt{2}x_1x_2 \ x_2^2 \ \sqrt{2}x_1 \ \sqrt{2}x_2] \end{aligned}$$

- Thus, a kernel function *implicitly* maps data to a high-dimensional space (without the need to compute each $\phi(\mathbf{x})$ explicitly).

What Functions are Kernels?

- For some functions $K(\mathbf{x}_i, \mathbf{x}_j)$ checking that $K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$ can be cumbersome.
- Mercer's theorem:
 - ***Every semi-positive definite symmetric function is a kernel***
- Semi-positive definite symmetric functions correspond to a semi-positive definite symmetric Gram matrix:

K=

$K(\mathbf{x}_1, \mathbf{x}_1)$	$K(\mathbf{x}_1, \mathbf{x}_2)$	$K(\mathbf{x}_1, \mathbf{x}_3)$	...	$K(\mathbf{x}_1, \mathbf{x}_n)$
$K(\mathbf{x}_2, \mathbf{x}_1)$	$K(\mathbf{x}_2, \mathbf{x}_2)$	$K(\mathbf{x}_2, \mathbf{x}_3)$		$K(\mathbf{x}_2, \mathbf{x}_n)$
...	...	...	...	...
$K(\mathbf{x}_n, \mathbf{x}_1)$	$K(\mathbf{x}_n, \mathbf{x}_2)$	$K(\mathbf{x}_n, \mathbf{x}_3)$	...	$K(\mathbf{x}_n, \mathbf{x}_n)$

Examples of Kernel Functions

- Linear : $K(x_i, x_j) = x_i^T x_j$
 - Mapping $\phi: x \rightarrow \phi(x)$ where $\phi(x)$ is x itself
- Polynomial of power p : $K(x_i, x_j) = (1 + x_i^T x_j)^p$
 - Mapping $\phi: x \rightarrow \phi(x)$ where $\phi(x)$ has $\left(\frac{d+p}{p}\right)$ dimensions
- Gaussian (radial-basis function):
- $K(x_i, x_j) = e^{-\frac{\|x_i - x_j\|^2}{2\sigma^2}}$
 - Mapping $\phi: x \rightarrow \phi(x)$, where $\phi(x)$ is infinite-dimensional: every point is mapped to a function (Gaussian); combination of functions for support vectors is the separator.

Non-linear SVMs Mathematically



- Dual problem formulation:

Find $\alpha_1 \dots \alpha_n$ such that
 $Q(\alpha) = \sum \alpha_i - \frac{1}{2} \sum \sum \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j)$ is maximized and
(1) $\sum \alpha_i y_i = 0$
(2) $\alpha_i \geq 0$ for all α_i

- The solution is:

$$f(\mathbf{x}) = \sum \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}_j) + b$$

Optimization techniques for finding α_i 's remain the same!

SVM Applications

- SVMs were originally proposed by Boser, Guyon and Vapnik in 1992 and gained increasing popularity in late 1990s.
- SVMs are currently among the best performers for a number of classification tasks ranging from text to genomic data.
- SVMs can be applied to complex data types beyond feature vectors (e.g. graphs, sequences, relational data) by designing kernel functions for such data.
- SVM techniques have been extended to a number of tasks such as regression [Vapnik *et al.* '97], principal component analysis [Schölkopf *et al.* '99], etc.
- Most popular optimization algorithms for SVMs use *decomposition* to hill-climb over a subset of α_i 's at a time, e.g. SMO [Platt '99] and [Joachims '99]
- Tuning SVMs remains a black art: selecting a specific kernel and parameters is usually done in a try-and-see manner.

GAUSSIAN MIXTURE MODEL (GMM)

Home of the Bright, Land of the Brave
Di Sini Bermulanya Pintar, Tanah Tumpahnya Berani



www.um.edu.my



[universityofmalaya](https://www.facebook.com/universityofmalaya)



[unimalaya](https://www.instagram.com/unimalaya)



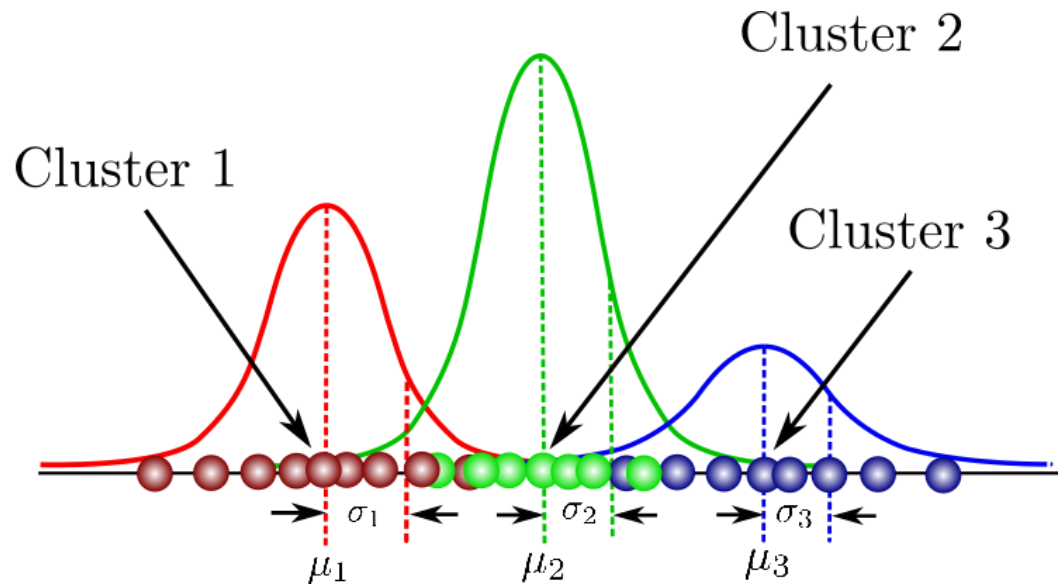
[uniofmalaya](https://www.youtube.com/uniofmalaya)



UNIVERSITI
MALAYA

Machine Learning Methods

Gaussian Mixture Model



Recap

Conditional probabilities:

$$p(A, B) = p(A|B)p(B) = p(B|A)p(A)$$

Bayes rule:

$$\overset{\text{Posterior Probability}}{p(A|B)} = \frac{\overset{\text{Conditional probability}}{p(A, B)}}{\underset{\text{Marginal Probability}}{p(B)}} = \frac{\overset{\text{Likelihood}}{p(B|A)} \overset{\text{Prior Probability}}{p(A)}}{p(B)}$$

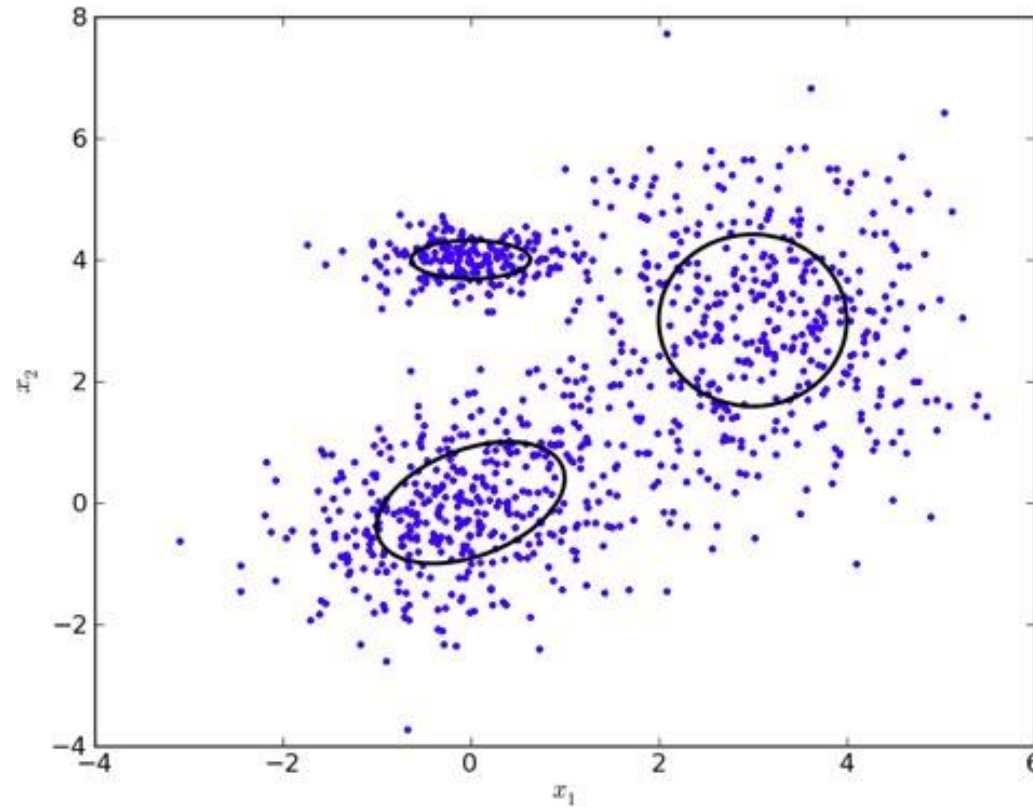
$$p(A = 1) = \sum_{i=1}^K p(A = 1, B_i) = \sum_{i=1}^K p(A|B_i) p(B_i)$$

	Tomorrow=Rainy	Tomorrow=Cold	P(Today)
Today=Rainy	$\frac{4}{9}$	$\frac{2}{9}$	$[\frac{4}{9} + \frac{2}{9}] = \frac{2}{3}$
Today=Cold	$\frac{2}{9}$	$\frac{1}{9}$	$[\frac{2}{9} + \frac{1}{9}] = \frac{1}{3}$
P(Tomorrow)	$[\frac{4}{9} + \frac{2}{9}] = \frac{2}{3}$	$[\frac{2}{9} + \frac{1}{9}] = \frac{1}{3}$	

$P(\text{Tomorrow} = \text{Rainy}) =$

Hard Clustering Can Be Difficult

- Hard Clustering: K-Means, Hierarchical Clustering, DBSCAN



Towards Soft Clustering

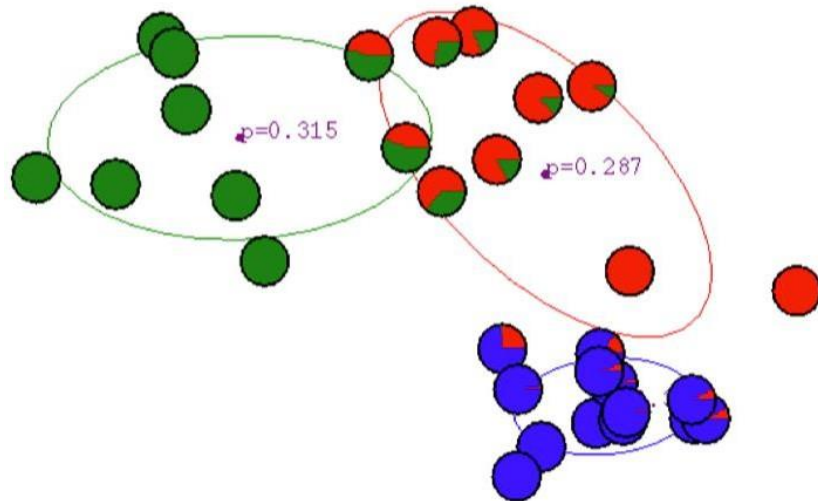
- **K-means**

- **hard assignment:** each object belongs to only one cluster

$$\theta_i \in \{\theta_1, \dots, \theta_K\}$$

- **Mixture modeling**

- **soft assignment:** probability that an object belongs to a cluster

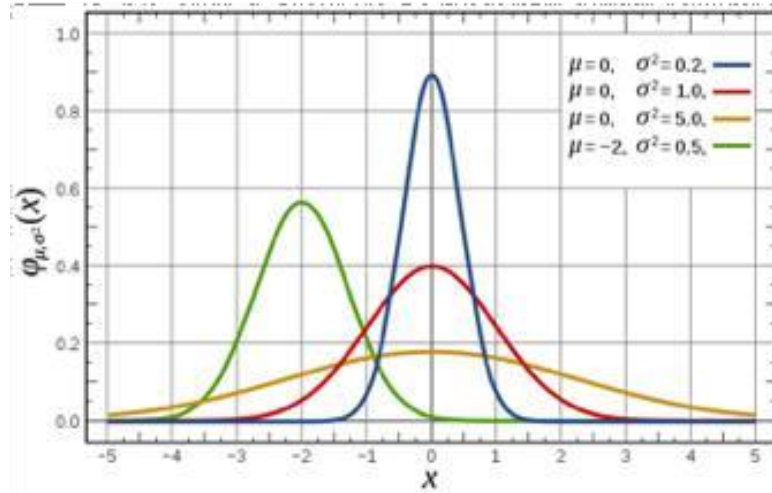


Gaussian Mixture Model

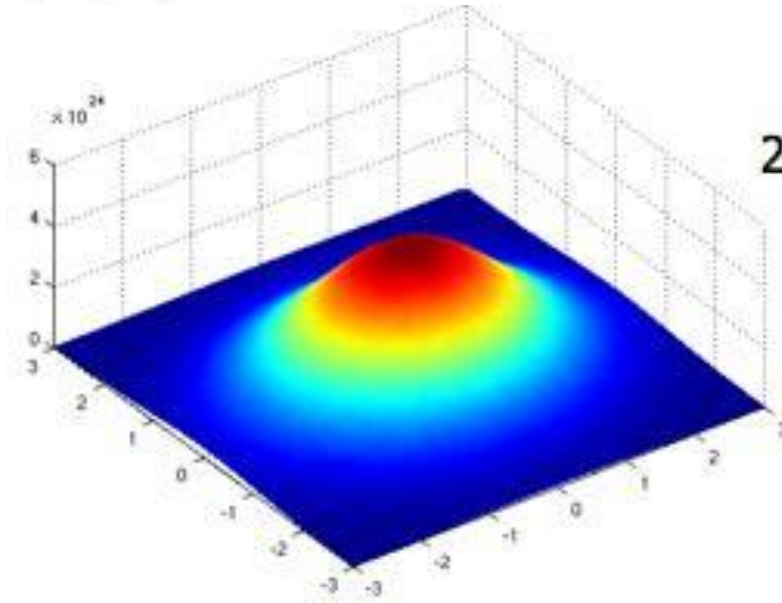
Watch [this](#) to get the concept of GMM before you go into the math in the slide.

Gaussian Distribution

1-d Gaussian



$$N(\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



2-d Gaussian

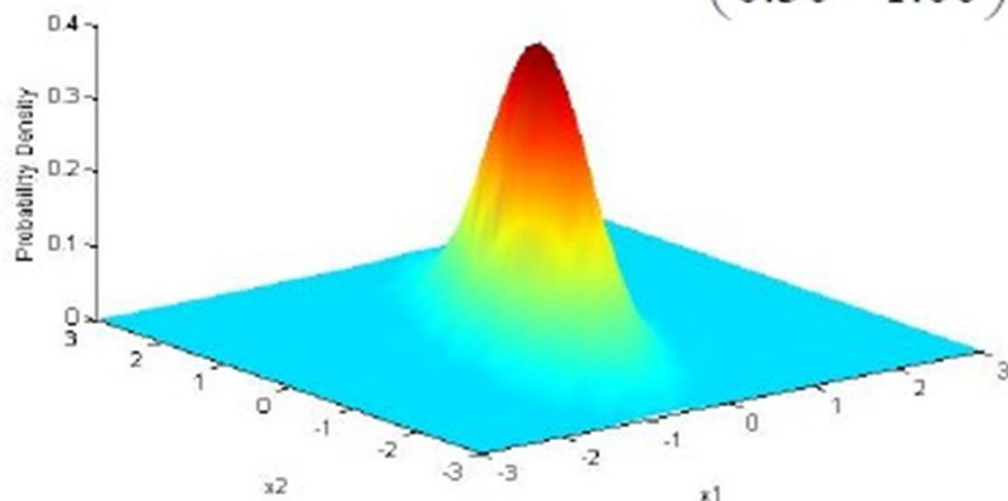
What is a Gaussian?

For d dimensions, the Gaussian distribution of a vector $x = (x^1, x^2, \dots, x^d)^T$ is defined by:

$$N(x | \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} \sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$$

where μ is the mean and Σ is the covariance matrix of the Gaussian.

Example: $\mu = (0,0)^T$ $\Sigma = \begin{pmatrix} 0.25 & 0.30 \\ 0.30 & 1.00 \end{pmatrix}$



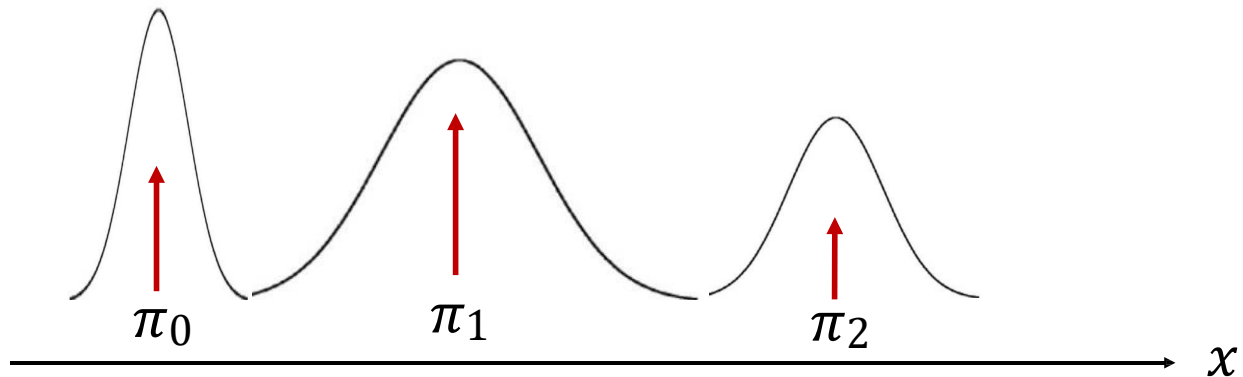
Mixture Models

- Formally a Mixture Model is the weighted sum of a number of pdfs where the weights are determined by a distribution, π

$$p(x) = \pi_0 f_0(x) + \pi_1 f_1(x) + \pi_2 f_2(x) + \dots + \pi_k f_k(x)$$

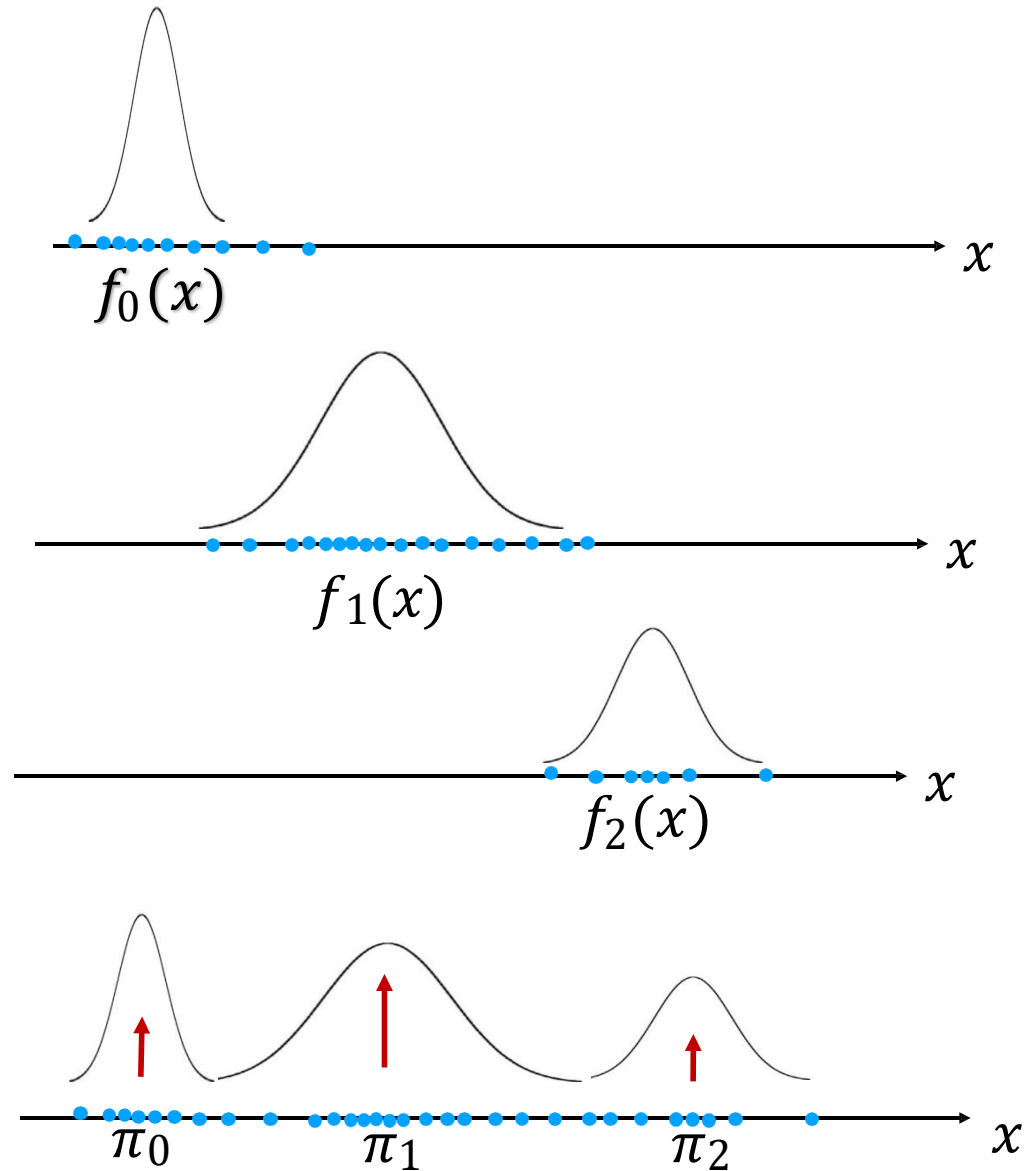
where $\sum_{i=0}^k \pi_i = 1$

$$p(x) = \sum_{i=0}^k \pi_i f_i(x)$$



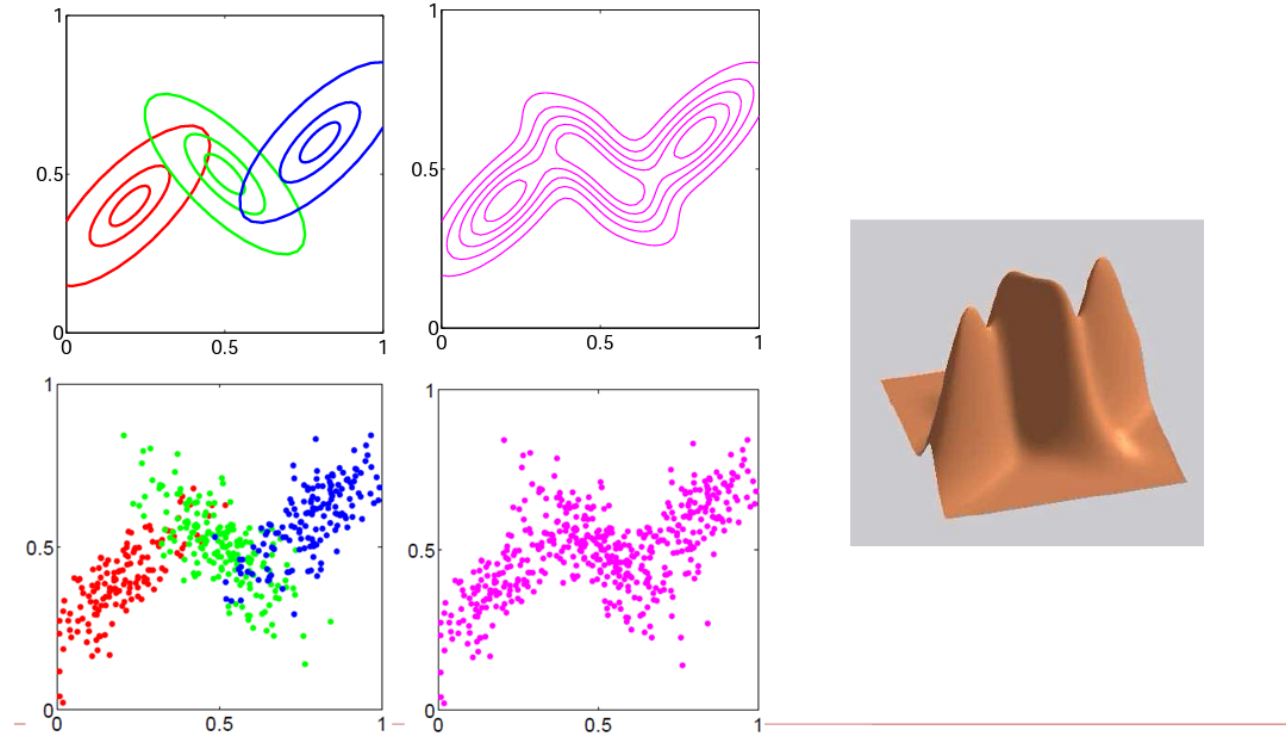
What is **f** in GMM?

$$p(x) = \pi_0 f_0(x) + \pi_1 f_1(x) + \pi_2 f_2(x)$$



Why GMM?

It creates a new pdf for us to generate random variables. It is a generative model.

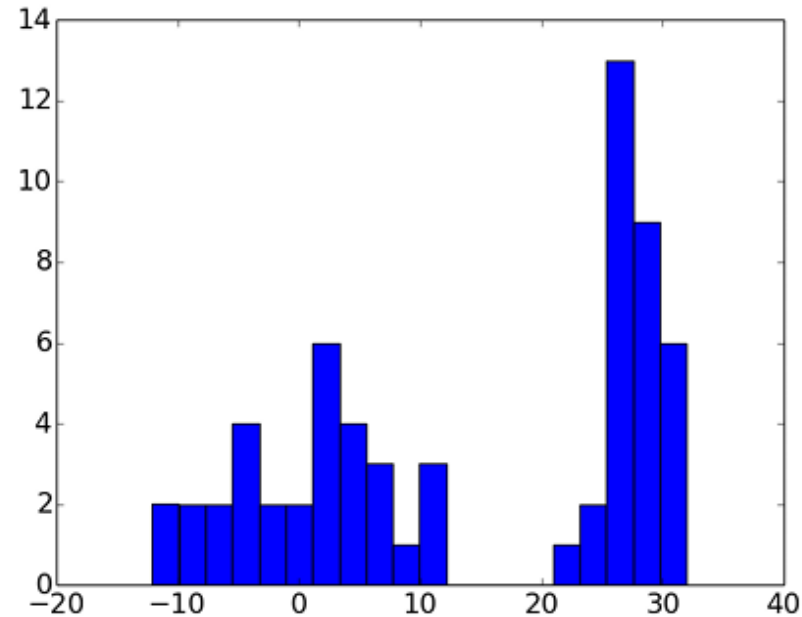


It clusters different components using a Gaussian distribution.

So it provides us the inferring opportunity. Soft assignment!!

Intuition

Histogram of temperature

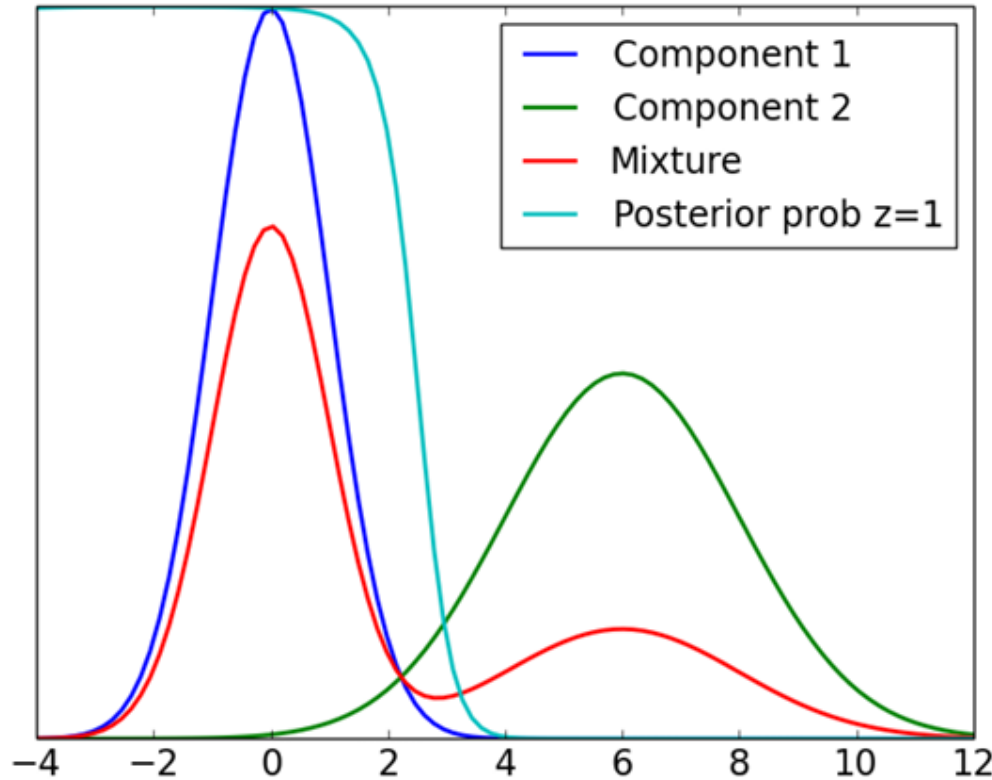


From the histogram, we see

1. Two distributions → could be from two different cities

Formulate into mathematics → observable: temperature,
latent variable, z : cities.

Gaussian Mixture Models



Write in mathematic notation

$$z \sim \text{Multinomial}(0.7, 0.3)$$

$$x \mid z = 1 \sim \text{Gaussian}(0, 1)$$

$$x \mid z = 2 \sim \text{Gaussian}(6, 2)$$

For general case,

$$z \sim \text{Multinoimal}(\boldsymbol{\pi})$$

$$x \mid z = k \sim \text{Gaussian}(\mu_k, \sigma_k).$$

$\boldsymbol{\pi}$ is a vector of probabilities (i.e. non negative values which sum to 1) known as the mixing proportions

Gaussian Mixture Models

In general, we can compute the probability density function (PDF) over $\mathbf{x}$ by **marginalizing out**, or summing out, z :

$$\begin{aligned} p(\mathbf{x}) &= \sum_z p(z) p(\mathbf{x} | z) \\ &= \sum_{k=1}^K \Pr(z = k) p(\mathbf{x} | z = k) \end{aligned}$$

In the previous example, this gives us:

$$p(x) = 0.7 \cdot \text{Gaussian}(0, 1) + 0.3 \cdot \text{Gaussian}(6, 2).$$

This PDF is a **convex combination**, or **weighted average** of the PDFs of the component distribution.

Posterior Inference

Assume we've chosen the parameters of model.

We want to infer, given a data point x , which component it is likely to belong to.

In mathematical term, we want to infer the posterior distribution $p(z|x)$.

We can infer the posterior distribution using Bayes' Rule:

$$p(z \mid \mathbf{x}) \propto p(z) p(\mathbf{x} \mid z).$$

Example I

Using the previous example, suppose we observe $x=2$ in the model described. We can compute the posterior probability $\Pr(z = 1|x)$

$$\Pr(z = 1) = 0.7$$

$$\Pr(z = 2) = 0.3$$

$$\begin{aligned} p(x | z = 1) &= \text{Gaussian}(2; 0, 1) \\ &\approx 0.054 \end{aligned}$$

$$\begin{aligned} p(x | z = 2) &= \text{Gaussian}(2; 6, 2) \\ &\approx 0.027 \end{aligned}$$

$$\begin{aligned} \Pr(z = 1 | x) &= \frac{\Pr(z = 1) p(x | z = 1)}{\Pr(z = 1) p(x | z = 1) + \Pr(z = 2) p(x | z = 2)} \\ &= \frac{0.7 \cdot 0.054}{0.7 \cdot 0.054 + 0.3 \cdot 0.027} \\ &\approx 0.824 \end{aligned}$$

Example II

Suppose our observations are two-dimensional, and we observe x_1 and want to make prediction about x_2 .

We can do this using posterior predictive distribution

$$p(x_2 | x_1) = \sum_z p(z | x_1) p(x_2 | z, x_1).$$

Example II

Consider if we have two-dimensional mixture of Gaussian model, where x_1 and x_2 are conditionally independent given z :

$$z \sim \text{Multinomial}(0.4, 0.6)$$

$$x_1 \mid z = 1 \sim \text{Gaussian}(0, 1)$$

$$x_2 \mid z = 1 \sim \text{Gaussian}(6, 1)$$

$$x_1 \mid z = 2 \sim \text{Gaussian}(6, 2)$$

$$x_2 \mid z = 2 \sim \text{Gaussian}(3, 2)$$

Suppose we observe $x_1=3$. We can compute the posterior distribution just like in the previous example:

$$\begin{aligned} \Pr(z = 1 \mid x_1) &= \frac{\Pr(z = 1) p(x \mid z = 1)}{\Pr(z = 1) p(x \mid z = 1) + \Pr(z = 2) p(x \mid z = 2)} \\ &= \frac{0.4 \cdot 0.0175}{0.4 \cdot 0.0175 + 0.6 \cdot 0.0431} \\ &= 0.213 \end{aligned}$$

Example II

Using the posterior, we can compute the posterior predictive distribution.

$$\begin{aligned} p(x_2 | x_1) &= \Pr(z = 1 | x_1) p(x_2 | z = 1) + \Pr(z = 2 | x_1) p(x_2 | z = 2) \\ &= 0.213 \cdot \text{Gaussian}(x_2; 6, 1) + 0.787 \cdot \text{Gaussian}(x_2; 3, 2) \end{aligned}$$

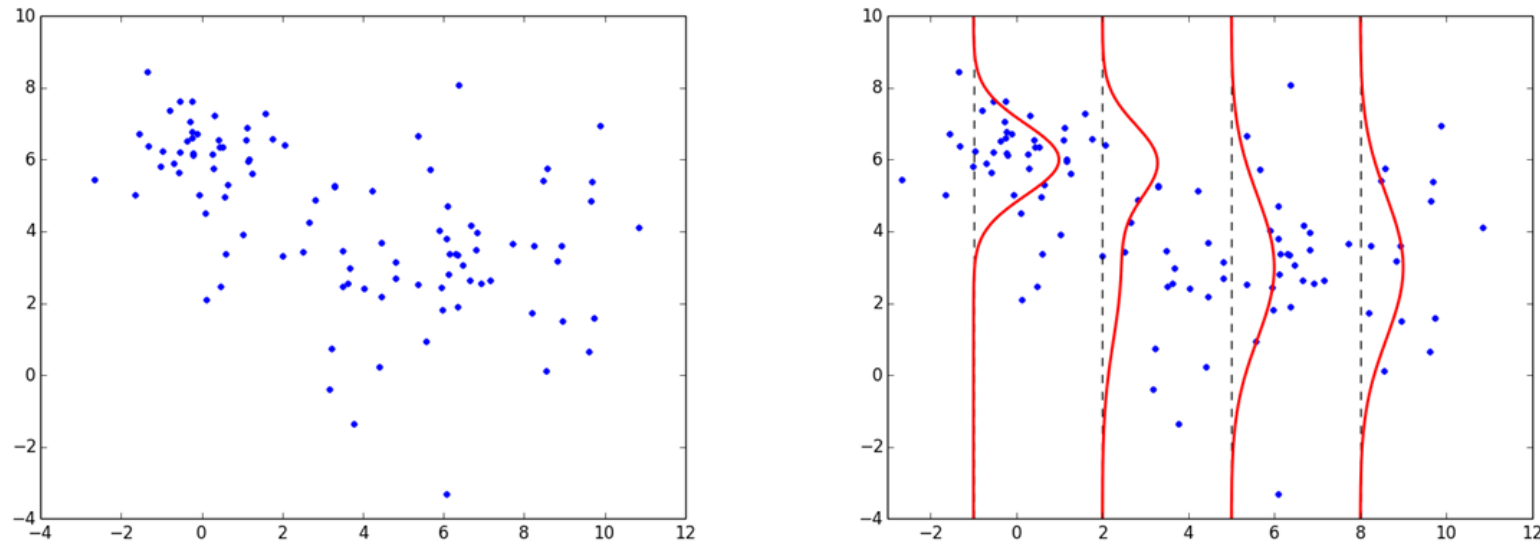


Figure 3: **Left:** Samples from the mixture of Gaussians model of Example 2. **Right:** The PDF of the posterior predictive distribution $p(x_2 | x_1)$, for various values of x_1 .

Learning

The parameters of GMM are:

- The mean μ_k and standard deviation σ_k associated with each component k . (Other kinds of mixture models will have other sets of parameters.)
- The **mixing proportions** π_k , defined as $\Pr(z = k)$.

Idea: The model do inferencing repeatedly to learn the parameters.

Learning

We use a method called “Maximum Likelihood Estimation” (MLE) to solve the problem.

Idea: The model do inferencing repeatedly to learn the parameters.

Learning

- Good idea to start by computing log-likelihood derivatives by setting the derivatives to 0 or we apply gradient descent.
- θ can be any of the parameters we want to learn, such as mixing proportion or the mean or standard deviation of one of the components.

$$\begin{aligned}\frac{d}{d\theta} \log p(\mathbf{x}) &= \frac{d}{d\theta} \log \sum_z p(z, \mathbf{x}) \\ &= \frac{\frac{d}{d\theta} \sum_z p(z, \mathbf{x})}{\sum_{z'} p(z', \mathbf{x})} \\ &= \frac{\sum_z \frac{d}{d\theta} p(z, \mathbf{x})}{\sum_{z'} p(z', \mathbf{x})} \\ &= \frac{\sum_z p(z, \mathbf{x}) \frac{d}{d\theta} \log p(z, \mathbf{x})}{\sum_{z'} p(z', \mathbf{x})}\end{aligned}$$
$$\begin{aligned}&= \sum_z \left(\frac{p(z, \mathbf{x})}{\sum_{z'} p(z', \mathbf{x})} \right) \frac{d}{d\theta} \log p(z, \mathbf{x}) \\ &= \sum_z p(z | \mathbf{x}) \frac{d}{d\theta} \log p(z, \mathbf{x}) \\ &= \mathbb{E}_{p(z | \mathbf{x})} \left[\frac{d}{d\theta} \log p(z, \mathbf{x}) \right]\end{aligned}$$

Expected derivative of
the joint log probability

Learning

- Optimization of means.

$$\ln p(x|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(x_n | \mu_k, \Sigma_k) \right\}$$

$$\begin{aligned} \frac{\partial \ln p(x|\pi, \mu, \Sigma)}{\partial \mu_k} &= \sum_{n=1}^N \frac{\pi_k N(x_n | \mu_k, \Sigma_k)}{\sum_j \pi_j N(x_n | \mu_j, \Sigma_j)} \Sigma_k^{-1} (x_k - \mu_k) = 0 \\ &= \sum_{n=1}^N \tau(z_{nk}) \Sigma_k^{-1} (x_k - \mu_k) = 0 \end{aligned}$$

$$\mu_k = \frac{\sum_{n=1}^N \tau(z_{nk}) x_n}{\sum_{n=1}^N \tau(z_{nk})}$$

Learning

- Optimization of covariance

$$\ln p(x|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(x_n | \mu_k, \Sigma_k) \right\}$$

$$\Sigma_k = \frac{1}{\sum_{n=1}^N \tau(z_{nk})} \sum_{n=1}^N \tau(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T$$

Learning

- Optimization of mixing term

$$\ln p(x|\pi, \mu, \Sigma) + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right)$$

$$0 = \sum_{n=1}^N \frac{N(x_n | \mu_k, \Sigma_k)}{\sum_j \pi_j N(x_n | \mu_j, \Sigma_j)} + \lambda$$

$$\pi_k = \frac{\sum_{n=1}^N \tau(z_{nk})}{N}$$

MLE of a GMM

$$\mu_k = \frac{\sum_{n=1}^N \tau(z_{nk}) x_n}{N_k}$$

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^N \tau(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T$$

$$\pi_k = \frac{N_k}{N}$$

$$N_k = \sum_{n=1}^N \tau(z_{nk})$$

Not a closed form solution!!
Use Expectation-Maximization Algorithm

Take-Home Messages

- The generative process of Gaussian Mixture Model
- Inferring cluster membership based on a learned GMM

THANK YOU

Home of the Bright, Land of the Brave
Di Sini Bermulanya Pintar, Tanah Tumpahnya Berani



www.um.edu.my



[universityofmalaya](https://www.facebook.com/universityofmalaya)



[unimalaya](https://www.instagram.com/unimalaya)



[uniofmalaya](https://www.youtube.com/uniofmalaya)



UNIVERSITI
MALAYA